How Does BMI Relate to Systolic Blood Pressure Across Age Groups? A Descriptive Analysis Using NHANES 2017-2018

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Repository: github.com/alilordifar/nhanes-bmi-blood-pressure-analysis

Introduction

Nearly half of U.S. adults have hypertension, and about 40% are classified as obese (NCHS, 2024), two conditions behind more than 900,000 cardiovascular deaths recorded in the country each year (AHA, 2025). These two risk factors frequently co-occur, but how does the association actually play out in a large national survey? This project focuses on a descriptive question: what is the relationship between body mass index and systolic blood pressure? We focus on systolic pressure specifically because it tends to be the stronger predictor of cardiovascular risk, especially as people age. Age itself may shape the BMI-blood pressure relationship too since arteries stiffen over time, though we don't try to isolate that here. We are not making a causal claim. We describe the strength and shape of the association between these variables in a large public health dataset. Excess body weight is often associated with hypertension, and a clear descriptive analysis can show how blood pressure tends to vary across individuals with different BMI levels. A descriptive picture like this matters because it shows which groups tend to have higher blood pressure and where the association is strongest.

Description of Data Source

We use data from the National Health and Nutrition Examination Survey (NHANES), the CDC's ongoing study designed to assess the health of the non-institutionalized civilian population of the United States. The survey combines in-home interviews with standardized physical exams and lab work conducted at mobile examination centers. Each row in the dataset is one participant. We pull four public 2017 to 2018 files: demographics (DEMO_J), body measures (BMX_J), blood pressure (BPX_J), and smoking (SMQ_J). BMI came from measured height and weight during the exam, and blood pressure was taken in a seated position after five minutes of rest, with up to three consecutive readings averaged, so neither variable relies on self-report. NHANES uses a multistage probability design that oversamples Black and Hispanic adults, older adults, and low-income individuals, and it surveys participants of all ages, though blood pressure exams are limited to those eight and older. We stay with the 2017 to 2018 cycle because it is the last pre-COVID release with complete blood pressure exam data. The full cycle includes about 9,254 participants; 5,176 remain after our merge and complete-case screen. Because we do not apply survey weights, our regression coefficients describe patterns within this analytic sample and should not be interpreted as nationally representative estimates.

Data Wrangling

We joined the four NHANES files on respondent ID (`SEQN`). That gave us 6,401 people with records in all four sources. We averaged the available systolic readings, kept BMI and age in their original units, coded sex as a binary indicator, recoded race/ethnicity to six groups, and recoded smoking to never, former, or current. Then we used complete case analysis. Of the 6,401 merged observations, 346 were missing blood pressure, 107 missing BMI, and 868 missing smoking status; 96 individuals were missing more than one variable, so the unique count dropped was 1225, leaving 5,176 complete cases. Most of the blood pressure and BMI missingness came from people who did not complete the exam. We then split the complete cases at random into 1,552 exploration observations and 3,624 confirmation observations.

Operationalization

Our outcome is systolic blood pressure, averaged across the three exam readings (`BPXSY1`, `BPXSY2`, `BPXSY3`) to smooth out white coat spikes, which is standard clinical protocol. We picked systolic over diastolic or mean arterial pressure because the literature ties it more consistently to cardiovascular risk. The main predictor is `BMXBMI`. We believe that waist circumference or DXA body fat would capture adiposity better, but BMI is what every prior study reports, so using it keeps our results comparable. Age (`RIDAGEYR`) and sex (`RIAGENDR`) enter the model directly, age as continuous to avoid throwing away information through binning. We include race/ethnicity (`RIDRETH3`) not as a biological variable but as a proxy for the socioeconomic and environmental factors that track with both BMI and blood pressure. Smoking status is operationalized two ways. Models 2 through 4 use a binary ever-smoker indicator from `SMQ020` (smoked at least 100 cigarettes in lifetime, yes or no) to stay consistent with the team's earlier specifications. Model 5 upgrades to a three-level factor combining `SMQ020` and `SMQ040` into never, former, and current smoker, because former and current smokers sit at different points on the cardiovascular risk spectrum and the binary version masked a significant difference between them. After merging on `SEQN` and dropping 346 observations missing blood pressure, 107 missing BMI, and 868 missing smoking, we retain 5,176 complete cases. A few limitations are worth flagging: BMI conflates muscle and fat, smoking is self reported with no pack-year detail, and all blood pressure readings come from a single visit, so we are working with a snapshot of something that fluctuates throughout the day.

Data Visualization

Figure 1a plots BMI against systolic blood pressure, colored by age group. The younger groups have visibly steeper smoothers than the oldest group, which is why we test a BMI by age interaction later in the report. Figure 1b overlays the fitted mean from Model 5 for a median age White male never smoker. The line rises with BMI, but the point cloud stays wide at every part of the x axis, so the figure also makes clear that BMI explains only a small share of the total variation.

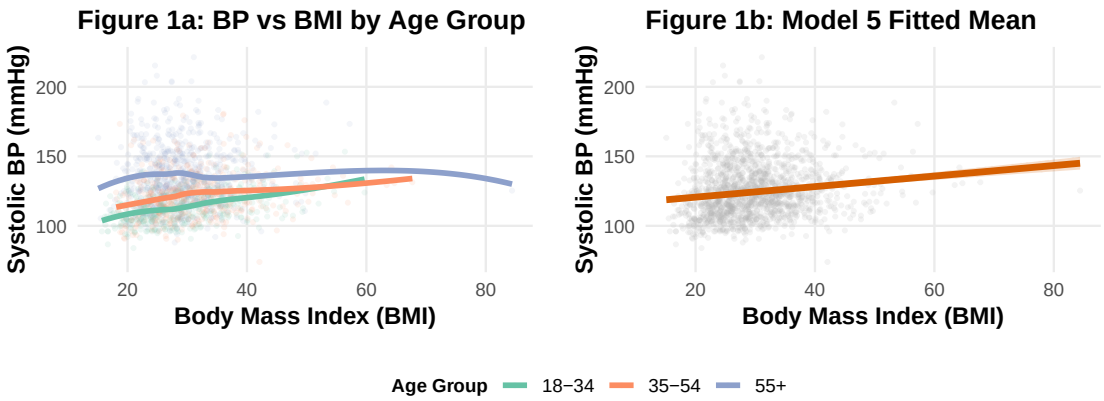


Figure 1a: BMI and systolic blood pressure colored by age group (exploration set, n = 1552). The younger age groups have steeper loess lines than the oldest group, which suggests that the BMI slope changes with age.

Figure 1b: BMI and systolic blood pressure (exploration set) with the Model 5 fitted mean and 95% confidence band overlaid for a median-age White male who never smoked. The fitted mean slopes upward, but the cloud around it remains wide.

Model Specification

Table 1 shows the five specifications we use, from BMI alone to the full interaction model with age squared, smoking category, sex, and race/ethnicity. The point of the sequence is to watch what happens to the BMI slope as we add basic controls, a curvature check, and an age interaction.

	Dependent variable:				
	Systolic Blood Pressure (mmHg)				
	(1)	(2)	(3)	(4)	(5)
BMI	0.45*** (0.04)	0.40*** (0.04)	−0.09 (0.22)	0.87*** (0.09)	0.89*** (0.09)
BMI^2			0.01** (0.003)		
Age		0.52*** (0.02)	0.52*** (0.02)	0.80*** (0.06)	0.37*** (0.10)
Smoking (binary)		−0.81 (0.60)	−0.87 (0.60)	−0.90 (0.60)	
Former Smoker					−1.90** (0.76)
Current Smoker					0.56 (0.83)
Sex (Female)					−2.11*** (0.57)
Race: Mex American					2.60*** (0.84)
Race: Other Hispanic					2.92** (1.13)
Race: Black					7.59*** (0.80)
Race: Asian					3.33*** (0.88)
Race: Other/Multi					3.52*** (1.27)
Age^2					0.005*** (0.001)
BMI × Age				−0.01*** (0.002)	−0.01*** (0.002)
Constant	112.84*** (1.37)	88.98*** (1.27)	96.67*** (3.53)	75.81*** (2.73)	81.70*** (2.93)
Observations	3,624	3,624	3,624	3,624	3,624
R^2	0.03	0.25	0.26	0.26	0.29
Adjusted R^2	0.03	0.25	0.25	0.26	0.28
Note:					*p<0.1; **p<0.05; ***p<0.01

Model Assumptions

We evaluate five key OLS assumptions below. Residual diagnostics look similar across Models 2 through 5, so adding polynomial, interaction, and demographic terms does not seem to introduce new violations.

1. **IID (Independent and Identically Distributed).** NHANES uses a multi-stage clustered design: it first samples primary sampling units (PSUs), then households within PSUs, then individuals within households. This structure means observations within the same PSU are not independent because residents share local environmental, economic, and healthcare access factors that jointly influence blood pressure. Oversampling

of certain racial/ethnic groups further means the observations are not identically distributed. Together, these features violate the IID assumption in a way we cannot fix post hoc without the survey design variables. The practical consequence is twofold. First, because our OLS standard errors ignore within-cluster correlation, they are almost certainly too small, making confidence intervals too narrow and p-values too optimistic. Second, because we do not apply sampling weights, point estimates reflect the composition of the analytic sample rather than the U.S. adult population that NHANES is designed to represent. A survey-weighted analysis such as `svyglm` with the survey design variables would address both problems. We flag this as the most serious limitation of our analysis.

2. **Linear Conditional Expectation.** The residuals-vs-fitted plots for all five models (Appendix) show the loess smoother tracking close to zero across the fitted range, which is our main evidence for approximate linearity. The quadratic BMI term in Model 3 ($p = 0.016$) reaches statistical significance on the confirmation set, but the R^2 improvement is negligible (0.254 to 0.256), and the term was not significant during exploration. We treat the linearity assumption as approximately satisfied, but it does not prove that the relationship is perfectly linear. Some mild curvature is visible at the extremes, where the data are sparse, so we treat this assumption as approximate rather than fully satisfied. If the true relationship curves at extreme BMI values, our linear specification would produce biased predictions in those regions while remaining reasonable in the interior of the data.

3. **No Perfect Collinearity.** Model 5 includes BMI, Age, BMI $\times$ Age, and Age^2, which are mechanically correlated with their component terms. This multicollinearity inflates standard errors on individual coefficients, making it harder to isolate each term's contribution, but it does not constitute perfect collinearity and does not bias the coefficient estimates themselves. We are not entirely sure how much this matters in practice given our sample size, but joint F-tests on groups of related terms remain valid either way.

4. **Homoskedastic Errors.** The residuals-vs-fitted plots (Appendix) show a visible fan shape: residual spread is wider at higher fitted values across all five models. A Breusch-Pagan test confirms significant heteroskedasticity ($BP = 173.89$, $p < 0.001$). When errors are heteroskedastic, default OLS standard errors are biased (typically too small where variance is high), which throws off confidence intervals and hypothesis tests. We address this by reporting heteroskedasticity-robust (HC) standard errors throughout. Robust SEs correct the variance estimates so that inference is asymptotically valid, but they do not change the point estimates or eliminate the heteroskedasticity itself. We could improve efficiency further with weighted least squares, but robust SEs are the more common and conservative choice here.

5. **Normally Distributed Errors.** The residuals are right-skewed, as visible in the positive tail of the residuals-vs-fitted plots (Appendix). Strict normality therefore does not hold. But with $n = 3624$ observations, the central limit theorem ensures that coefficient estimates and their robust standard errors are approximately normal under mild regularity conditions. So we still trust the hypothesis tests for the slope coefficients. The main consequence of non-normality at this sample size is for prediction intervals rather than coefficient inference: individual-level prediction intervals based on a normal assumption will undercover in the right tail, where blood pressure values are higher than the symmetric interval expects. This concern is less consequential than the design-based independence issue above.

Model Results and Interpretation

All results below are estimated on the confirmation set (n = 3,624); the patterns mirror what we observed during exploration. The simple linear regression model shows a positive and statistically significant relationship between BMI and systolic blood pressure: a one-unit increase in BMI is associated with approximately 0.45 mmHg higher systolic pressure. The magnitude is small and the R-squared of 0.028 confirms that BMI alone explains very little of the variation in blood pressure.

When the model is extended to include age and smoking status, R-squared rises to 0.25. BMI remains statistically significant but its estimated effect decreases slightly, suggesting that part of the association in Model 1 is explained by age. Age shows a strong positive relationship with blood pressure and is the dominant predictor in this specification. Smoking status does not appear to have a statistically significant effect.

In Model 3, adding a BMI^2 term produces a statistically significant coefficient on the confirmation set ($p = 0.016$), though the term was not significant during exploration. The R^2 moves from 25.4% to 25.6%, a negligible improvement. Because the modeling decision to drop the quadratic term was made on the exploration set, and because the practical gain is minimal, we treat the BMI-blood pressure relationship as approximately linear and do not carry the quadratic term into later specifications.

In Model 4, including an interaction term between BMI and age shows that the effect of BMI on blood pressure depends on age. The interaction coefficient is negative and statistically significant, indicating that BMI has a stronger association with blood pressure at younger ages and a weaker association at older ages. While the overall model fit remains similar to Model 2, this specification provides insight on how BMI and age jointly relate to blood pressure.

In Model 5, adding a quadratic age term shows blood pressure climbing faster at older ages (Age^2 $p < 0.001$), and the BMI $\times$ Age interaction remains significant and negative, so BMI's link to blood pressure is weaker in older adults. We also swapped binary smoking for a three level classification (never, former, current) and added sex and race. Females have about 2 mmHg lower systolic pressure than males ($p < 0.01$), and Black participants have roughly 7 mmHg higher systolic pressure than White participants ($p < 0.01$). Since hypertension is defined as systolic BP $\geq$ 130 mmHg, a gap of nearly 7 mmHg can mean the difference between a normal reading and a Stage 1 diagnosis for two otherwise similar people. The three level smoking variable reveals that former smokers have slightly lower blood pressure than never smokers ($p = 0.009$), something the binary variable missed. We read that finding cautiously because it could reflect survivorship bias or lifestyle changes that come with quitting. The adjusted R^2 improves to 0.28. These are adjusted associations, not causal effects. Diet, medication, or physical activity could shift the numbers. Table 2 illustrates this with example profiles: for a 30 year old male never smoker, going from BMI 25 to 35 predicts a larger systolic increase than for a 60 year old with the same profile. The table also shows how baseline blood pressure shifts across racial groups even when BMI, age, sex, and smoking are held constant.

Table 1: *

Table 2: Predicted Systolic BP (mmHg) Under Model 5: Never-Smoker (Example Profiles)

Age	Sex	Race	BMI = 25	BMI = 35	Difference
30	Male	White	111.7	117.7	6
30	Male	Black	119.3	125.3	6
30	Male	Asian	115	121	6
30	Female	White	109.6	115.6	6
60	Male	White	127.6	130.6	3
60	Male	Black	135.1	138.1	3
60	Female	White	125.4	128.4	3

Overall Effect

We set out to describe the relationship between BMI and systolic blood pressure in U.S. adults. Across all five specifications the answer is consistent: higher BMI tracks with higher systolic pressure, the association is approximately linear (the quadratic BMI term, while significant on the confirmation set, added less than 0.2 percentage points to R^2), and it is strongest in younger adults. Age is the dominant predictor, race/ethnicity contributes a clinically meaningful gap (roughly 7 mmHg for Black versus White participants), and sex adds a modest offset, while smoking status matters only when coded at three levels. Together these covariates explain about 28% of systolic pressure variation, leaving sodium intake, physical activity, antihypertensive medication, and whatever else we did not measure to account for the rest. The data are cross sectional and we did not apply survey weights, so these associations describe this analytic sample and should not be read as causal or nationally representative. If future work linked these results to longitudinal NHANES panels or pulled in prescription records from the RX files, it could test whether the age dependent patterns hold up once treatment history is accounted for. The most notable pattern is that a 10 unit BMI difference in a younger adult corresponds to a predicted systolic pressure near the 130 mmHg hypertension threshold, suggesting that the BMI blood pressure association is most pronounced in the age range where it is least expected.

Appendix

Data Source

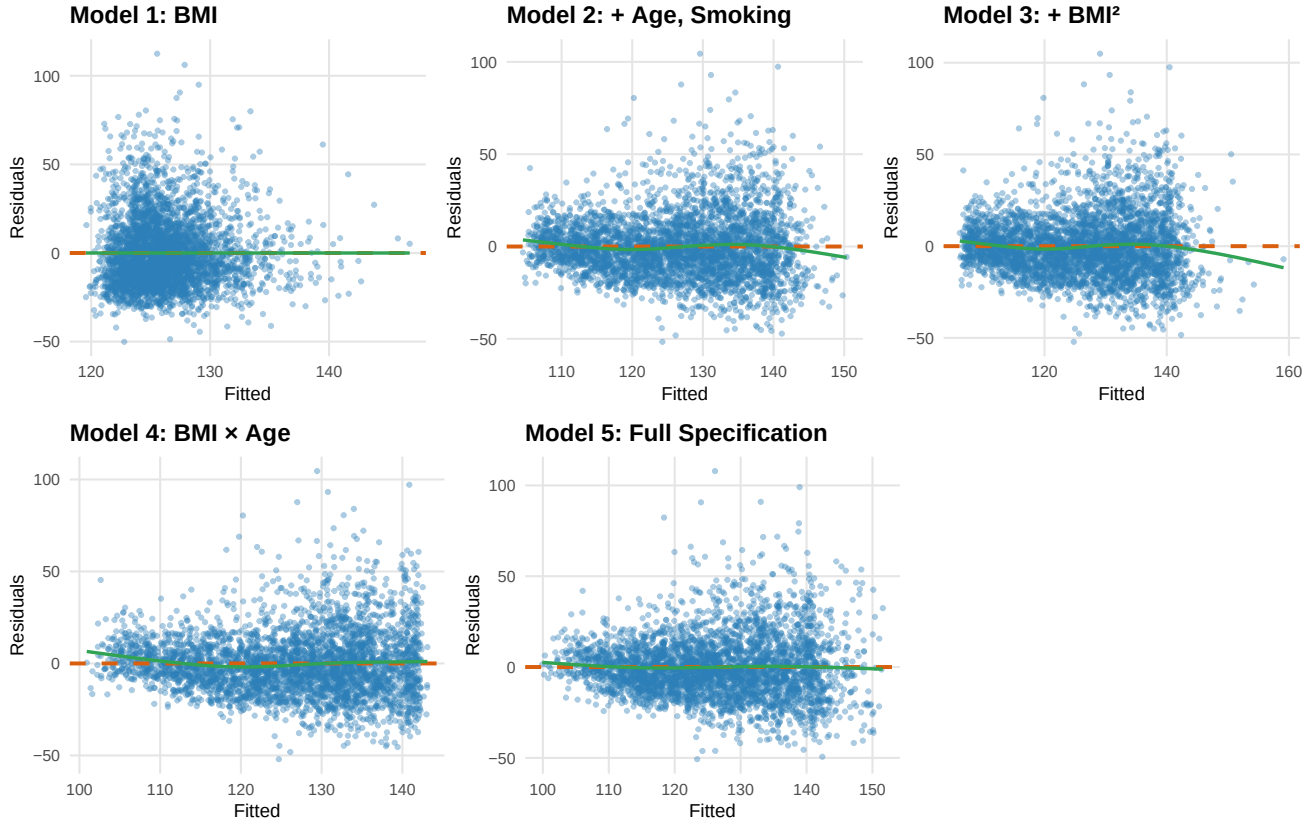
NHANES 2017-2018 data: https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2017

Model Specifications Tried

1. **Model 1** (bp ~ bmi): Baseline association. BMI is significant but explains only 2.8% of variance, which suggested that additional covariates were needed.
2. **Model 2** (bp ~ bmi + age + smoking): Adding age dramatically improved fit (R^2 from 0.028 to 0.25). Binary smoking was nonsignificant.
3. **Model 3** (bp ~ bmi + bmi^2 + age + smoking): Tested nonlinearity in BMI. The quadratic term was not significant during exploration; it reached significance on the confirmation set ($p = 0.016$) but with negligible R^2 improvement, so we retained the linear specification.
4. **Model 4** (bp ~ bmi × age + smoking): Interaction revealed BMI matters more for younger adults and was motivated by the diverging slopes in Figure 1.
5. **Model 5** (bp ~ bmi × age + age^2 + smoking3 + sex + race): Added demographic controls and 3-level smoking. Sex and race are significant; former smoker is now significant. Adj R^2 improved to 0.28.

We also explored physical activity, alcohol use, sodium intake, and BP medication as covariates. However, missingness in these variables (37–67%) reduced the sample from ~5,000 to 939 observations, introducing selection bias and collapsing the R^2 to 0.08 on the restricted subsample. These variables were therefore excluded from the final models.

Residuals-vs-Fitted Plots



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